Learning to Prefer: The Development of Abstract Binomial Ordering Preferences in Language Models

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Abstract

A defining property of human language is the ability to trade off between stored knowledge and abstract knowledge, yet what drives this trade-off remains poorly understood. In the present study, we investigate whether transformer language models trained on human-scale data exhibit a comparable trade-off, using binomial ordering preferences as a test case. We train six autoregressive language models from scratch (125M, 350M, and 1.3B parameters) on two corpora (BabyLM and a subset of C4) and checkpoint models throughout training to examine how ordering preferences emerge. We find that although all models' preferences are driven by their experience, smaller models show evidence of learned abstract preferences, while larger models show little such evidence at the final checkpoint; however trajectory analyses suggest they are sensitive to abstract preferences earlier in training. The present study demonstrates that model size and training data both play a role in the emergence of abstract knowledge in language models and further contributes to open-access research by making all models and checkpoints publicly available.

1 Introduction

The ability to trade off between stored and abstract knowledge is central to human language. A proficient language user must be able to apply abstract rules productively (for example, generating the plural of a novel word like *wug* as *wugs*) while simultaneously drawing on stored knowledge that resists those rules (knowing that the plural of *goose* is *geese*, not *gooses*). There is substantial evidence that humans navigate this trade-off dynamically as a function of linguistic experience (e.g., Houghton, 2025b; Morgan & Levy, 2016a; 2024), yet the mechanisms driving it remain poorly understood.

In the present study, we investigate whether transformer language models trained on human-scale data exhibit a comparable trade-off between stored and abstract knowledge, and whether they mirror what is observed in humans. Our main contributions are as follows:

- We train six autoregressive language models from scratch on two corpora (BabyLM, Charpentier et al., 2025; and a subset of C4, Raffel et al., 2020) at three model sizes (125M, 350M, and 1.3B parameters), allowing us to examine how training data and scale jointly shape ordering preferences. In order to promote open access research, we also make these 6 models (along with all checkpoints) publicly accessible.
- We replicate and extend Houghton et al. (2025) by examining whether models trained on a developmentally plausible amount of data show evidence of abstract preferences, rather than solely memorization.
- We examine the full learning trajectory of binomial ordering preferences across training allowing us to assess not only whether human-like preferences emerge, but when and how they develop.
- We introduce a preference deviation measure that captures the deviation between a model's ordering preference and what training statistics alone would predict, enabling a more direct test of whether model preferences are driven by factors beyond memorization.

1.1 Abstract Knowledge and Stored Knowledge in Humans

There is strong evidence for both abstract and stored knowledge in humans, with abstract knowledge being considerably less controversial (cf. Ambridge, 2020). For example, in a seminal paper, Berko (1958) demonstrated that children reliably apply morphological rules to novel words they have never encountered, yielding the now-famous example of a creature called a *wug*.

Evidence for abstract knowledge spans virtually every level of linguistic representation, from syntactic structures (evidenced by syntactic priming, Bock, 1986), to phonological representations (evidenced by phonological priming, Goldinger, 1992), to semantic representations (Meyer & Schvaneveldt, 1971). Priming effects provide compelling evidence for abstraction: for one context to facilitate processing in another, the two must share some underlying representation. This elegantly accounts for why, for example, sentences with entirely different words but a shared syntactic structure can prime one another.

Abstract knowledge also shapes the learning and processing of specific constructions. For example, Benor & Levy (2006) demonstrated that binomial orderings in corpus data (e.g., *bishops and seamstresses* vs. *seamstresses and bishops*) are driven by a range of phonological, semantic, and metrical constraints, such as a preference to place the more powerful or shorter word first. Morgan & Levy (2016a) operationalized these constraints by coding a set of binomials for these features and fitting a logistic regression model to corpus data, yielding a single fitted value per binomial (abstract preference) that summarizes the degree to which abstract constraints favor one ordering over the other. They found that abstract preferences predicted human ordering preferences for low-frequency binomials, while relative frequency (the proportion that a binomial appears in corpus data in alphabetical ordering relative to nonalphabetical ordering) predicted the ordering of high-frequency items. A follow-up study (Morgan & Levy, 2024) subsequently showed that while relative frequency remains a strong predictor for high-frequency binomials, abstract preferences exert an influence across the full frequency spectrum. Together, these results suggest that humans draw on stored knowledge for high-frequency binomials while relying on abstract preferences for lower-frequency items.

The trade-off between abstract and stored knowledge as a function of language usage has a long history in the literature (e.g., Bybee, 1999; Houghton, 2025b; Kapatsinski, 2010; Kapatsinski & Radicke, 2009; Morgan & Levy, 2015; 2016a;b; Stemberger & MacWhinney, 2004; 1986). For example, Stemberger & MacWhinney (2004) demonstrated that inflection errors are less common for high-frequency words than low-frequency words. Similarly, Bybee (1999) showed that *don't* is more phonetically reduced in high-frequency phrases (e.g., *I don't know*) than in low-frequency ones (e.g., *I don't go*). Processing measures have provided convergent evidence: recognition of constituent words is harder in high-frequency (Kapatsinski & Radicke, 2009) and high-predictability (Houghton, 2025b) phrases (e.g., recognizing *up* in *pick up*), consistent with holistic storage of these sequences. Despite this broad empirical support, however, the mechanism driving the trade-off remains unclear.

1.2 Accounts of Abstract Knowledge and Stored Knowledge

One influential mechanistic account appeals to neural network models, which offer an explanation for how abstraction and storage can emerge from the same underlying system: by learning to map inputs onto distributional representations that reflect both the statistical regularities shared across items and the idiosyncratic properties of individual items, these systems can in principle capture both rule-like generalizations and item-specific stored knowledge within a single architecture. Neural networks enable this ability to capture both types of knowledge through hidden layers, which project the input into a new representational space where previously intractable classification boundaries become linearly separable (see Appendix Section A for a concrete illustration).

Multi-layer neural networks underlie a great deal of success behind the transformer architecture (Vaswani et al., 2017), which is the architecture behind nearly every successful language model that has emerged within the last few years. However, it is unclear whether transformer models trade off between stored and abstract knowledge in a way that is similar

to humans. Indeed, the evidence for abstract knowledge altogether is quite mixed (Haley, 2020; Houghton et al., 2025; Jian & Manning, 2026; Kapatsinski, 2026; Lasri et al., 2022; Li & Wisniewski, 2021; Li et al., 2023; McCoy et al., 2023; Misra & Mahowald, 2024; Yao et al., 2025).

On one hand, there is evidence that GPT-2 (Radford et al., 2019) can generate both novel words and novel syntactic structures (McCoy et al., 2023), and that models can generalize learned knowledge beyond the specific constructions in which they encounter it (Misra & Mahowald, 2024). For example, Misra & Mahowald (2024) examined whether models can learn the rare Article Adjective Numeral Noun construction (e.g., *a beautiful five days*) without direct exposure, and found that models can learn it by generalizing from similar structures they have encountered, such as *a few days*. Similarly, Kapatsinski (2026) demonstrated that GPT-2 exhibits adaptive partial pooling during training, such that predictions become increasingly context-specific for frequent contexts while continuing to draw on broader distributional information for rare ones, a pattern that closely mirrors the frequency-dependent trade-off between stored and abstract knowledge observed in humans. Relatedly, Jian & Manning (2026) examined whether GPT-2's learning trajectory over the course of training is better characterized by abstraction-first or exemplar-first accounts. Using a novel divergence-based metric, they tracked how next-word prediction distributions change across training checkpoints. Under an exemplar-first account, next-word predictions for individual items should initially diverge from others in the same class before converging as training progresses. Instead, they found that class-level distinctions arise as soon as any item-specific change is detectable, suggesting that transformer models have an inductive bias toward abstraction when learning language.

On the other hand, there is evidence that these models rely heavily on memorization (Houghton et al., 2025; McCoy et al., 2023). Although McCoy et al. (2023) demonstrated that GPT-2 can generate novel words and syntactic structures, the same study found that it also reproduces a substantial portion of its training data verbatim. Similarly, Houghton et al. (2025) examined binomial ordering preferences in eight models ranging from 124M to 70B parameters and found no evidence of abstract preferences in any of them: ordering preferences were driven entirely by relative frequency across the board. Critically, a key ambiguity in this study is that the models were trained on trillions of tokens, making it impossible to determine whether the absence of abstract ordering preferences reflects a genuine architectural limitation or simply the overwhelming influence of having encountered each binomial many times over. Whether transformer models can trade off between abstraction and storage in a human-like manner therefore remains an open question.

A further complication is that examining models only at their final training checkpoint may obscure important dynamics. If models initially rely on abstract knowledge and gradually shift toward stored representations as training progresses, end-of-training snapshots would miss this trajectory entirely. Indeed, Houghton et al. (2025) explicitly flagged the absence of such an analysis as a limitation, noting that their study was not designed to investigate the effect of training data scale or how ordering preferences develop across training. Examining learning curves across checkpoints is therefore essential for understanding not only whether human-like abstract-storage trade-offs emerge, but when and how they develop.

1.3 Present Study

To examine whether language models can trade off between abstract and stored knowledge in a human-like manner, we extend Houghton et al. (2025) to models trained on a developmentally plausible amount of data. Specifically, we train six autoregressive language models from scratch using the OPT architecture (Zhang et al., 2022), crossing two training corpora (BabyLM, Charpentier et al., 2025; and a subset of C4, Raffel et al., 2020) with three model sizes (125M, 350M, and 1.3B parameters). Models are checkpointed throughout training to track how binomial ordering preferences evolve as a function of experience.

For each model and checkpoint, we computed the model's ordering preference along with the model's preference deviation which captures the degree to which the model's preferences deviate from what relative frequency alone would predict. We then examine

how these change as a function of model's experience with binomials across training. By examining ordering preferences across training, we assess not only whether this human-like abstract-storage trade-off emerges, but when and how it develops.

2 Model Training

In order to examine the representation of binomials in language models, we trained six autoregressive language models from scratch using the OPT architecture (Zhang et al., 2022), crossing two training corpora with three model sizes (125M, 350M, and 1.3B parameters), for a total of six models. All models share a common tokenizer: a Byte-Pair Encoding (BPE) tokenizer with a vocabulary of 8,192 tokens and a context window of 1,024 tokens, trained once on the full BabyLM corpus and reused across all six models to ensure that differences between models reflect training data and scale rather than tokenization. Models were trained on two NVIDIA H100 GPUs using PyTorch Distributed Data Parallel (DDP) in bfloat16 precision, with AdamW optimization and a linear learning rate schedule with 4,000 warmup steps.

The first corpus that we trained the models on was BabyLM (Version 3, Charpentier et al., 2025), a developmentally plausible corpus of approximately 150M tokens. Each model was trained for 64 epochs over this corpus, yielding approximately 9.1B total training tokens. The second corpus was a 10B-token subset of C4 (Raffel et al., 2020), a filtered English web text corpus. C4 models were trained for a single pass over this corpus, yielding approximately 10B training tokens. Comparing the two corpora allows us to examine how the content and diversity of the training data shapes the learning of binomials in language models.

Checkpoints were saved approximately every 10M tokens throughout training and pushed to HuggingFace Hub, giving us dense coverage of each model's learning trajectory. All models and checkpoints will be made publicly available.[1] Key training parameters for each model are summarized in the Appendix (Table 2).

All six models showed smooth convergence in training loss throughout training, starting near the theoretical random baseline ($\ln 8192 \approx 9.01$) and decreasing consistently. Perplexity on the held-out validation set confirmed successful training for all models; full loss curves and perplexity figures are reported in the Appendix section C. Notably, the BabyLM 1.3B model showed elevated final validation perplexity relative to the 350M model (2,557 vs. 290), consistent with mild overfitting given the small corpus (approximately 150M tokens) — unsurprising for a 1.3B parameter model trained on such limited data.

3 Binomial Ordering Preferences

3.1 Dataset

The dataset used in all analyses is taken from Morgan & Levy (2015), which comprises 594 attested English binomial expressions annotated for various phonological, semantic, and metrical constraints identified by Benor & Levy (2006). For each binomial, the corpus provides an abstract preference score (AbsPref), derived from a logistic regression model combining these phonological, semantic, and metric constraints (Morgan & Levy, 2016a). Abstract Preferences capture the degree to which these constraints favor one ordering over the other independently of corpus experience. This is the same set of items used in Houghton et al. (2025).

Frequency statistics for each binomial were estimated directly from the two training corpora rather than from an external source. Crucially, training from scratch with intermediate checkpoints allows us to compute the exact counts for all binomial orderings at every stage of training, for each model separately, rather than relying on approximations derived from corpus statistics. For each binomial and checkpoint, we computed the total number of

[1] The huggingface account would reveal the identity of one of the authors and is thus omitted from the double-blind review phase.

Statistic	BabyLM	C4
Total attested binomials	594	594
Both orderings attested ($n > 0$)	185	553
Log overall frequency: mean	6.95	5.46
Log overall frequency: SD	1.47	2.19

Table 1: Summary statistics for attested binomials in each training corpus. *Both orderings attested* is the number of binomials for which both the alphabetical and non-alphabetical orderings appear at least once in the corpus. Log overall frequency statistics are calculated after applying this exclusion.

exposures across both orderings. We eliminated binomials that occurred 0 times in both orderings.[2] Note that this resulted in a different number of items for the babyLM analyses vs the C4 analyses given the differences in binomials present in each corpus. Table 1 summarizes the coverage of the attested binomials in each corpus.

3.2 Analyses

In this section we present two complementary analyses examining how our models represent binomial ordering preferences over the course of training. The first examines model **preference** and **preference deviation** at the end of training, asking what factors predict each measure at the final checkpoint. The second examines how these preferences develop over the course of training, tracking how the influence of abstract and frequency-based factors changes across checkpoints and how the overall magnitude of preferences evolves as training progresses.

3.2.1 Ordering Preferences at the Final Checkpoint

We consider two complementary outcome measures. The first is model **preference**: for each model, we computed the preference for a given binomial by prepending each of 51 diverse prompt prefixes to both orderings, computing the difference in summed token log-probability between the alphabetically ordered form (e.g., *bread and butter*) and the non-alphabetically ordered form (e.g., *butter and bread*) within each prompt context, and averaging those per-prompt differences. Positive values indicate a stronger preference for the alphabetically ordered form (see Section D.1 for a formal definition). The second outcome is the **preference deviation**: the difference between the model's preference score and the log relative frequency ratio for that binomial (see Section D.2 for a formal definition). Because we train our models from scratch and checkpoint them throughout training, we can compute the exact number of times each model has seen each ordering of each binomial at every checkpoint. A preference deviation of zero means the model's preference exactly matches what relative frequency predicts; positive values indicate a stronger preference for the alpha-ordered form than corpus statistics alone would suggest, and negative values the reverse.

To help build an intuition for these measurements, Figure 1 illustrates both measures for five example binomials from the C4 125M model at the final checkpoint. For example, *crannies and nooks* is strongly preferred in the non-alphabetical ordering (i.e., *nooks and crannies* is strongly preferred), while *bride and groom* is strongly preferred in the alphabetical ordering. Moreover, *nooks and crannies* is preferred even more strongly than would be expected by the the model's training data, while there is little difference between the expected preference and model's preference for the binomial *authority and force*. Interestingly, Figure 1 demonstrates that there can be quite large difference between the training data and the model's preference. As a case in point, while the expected preference of *heir and son* and *crannies and nooks* is quite similar, the model learns a rather extreme preference for the non-alphabetical ordering of *crannies and nooks*, but has very little preference for *heir and son* in either ordering (despite

[2]This yielded significantly fewer binomials in the BabyLM corpus due to it being a small corpus.

heir and son occurring more than 300 times more in the nonalphabetical ordering (*son and heir*) than in the alphabetical ordering.

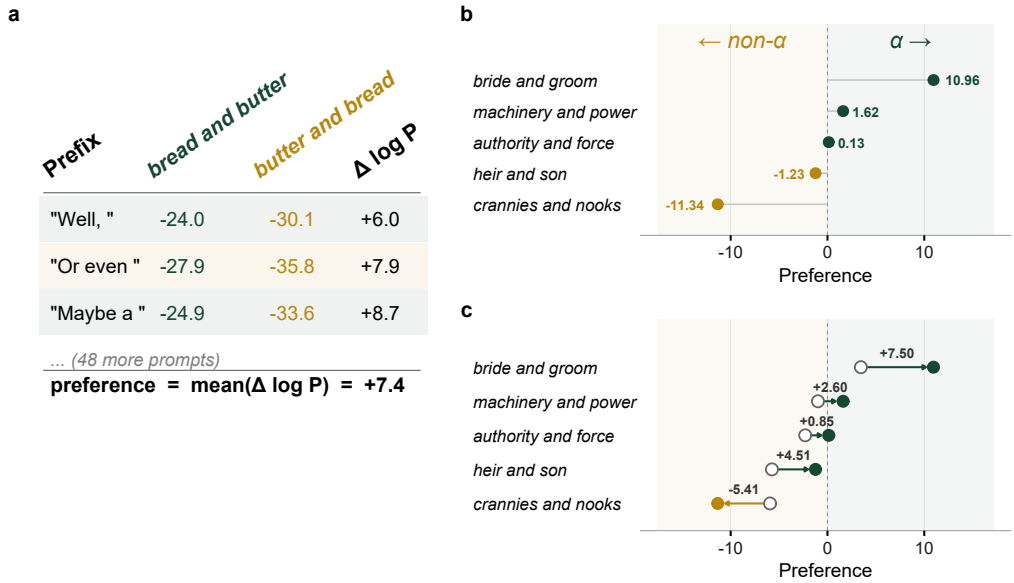

Figure 1: Illustration of our outcome variables. **(a)** For each binomial, 51 diverse prompt prefixes are prepended to both orderings; per-prompt log-probability differences ($\Delta \log P$) are averaged to yield a single preference score. **(b)** Example preference scores. Positive values indicate preference for the alphabetically ordered form; negative values indicate preference for the reverse. **(c)** Preference deviation. Open circles show the frequency-based prediction (log relative frequency ratio); filled circles and arrows are coloured green if the model's preference exceeds the frequency prediction and gold if it falls below; the arrow connects the two points and the annotated value is the preference deviation. Values near zero indicate that the model's preference matches what training data predicts; positive values indicate stronger preference for the alphabetical order than training predicts.

For each outcome, we implemented a Bayesian mixed-effects regression model in R (R Core Team, 2025) using the brms package (Bürkner, 2017) with fixed effects for **AbsPref** (Abstract Preference), **RelFreq**, and **overall frequency**, interactions between each of AbsPref and RelFreq with overall frequency, and random intercepts for binomial.[3] AbsPref captures the degree to which abstract phonological and semantic constraints favor one ordering over the other (Morgan & Levy, 2016a); RelFreq is the proportion of corpus occurrences in the alphabetically ordered form, centered (see Section D.3); and overall frequency is the natural log of total occurrences in either ordering, logged and centered. The interaction terms allow us to test whether the influence of abstract constraints and relative frequency varies with overall binomial frequency. All models used uninformative priors.

$$\begin{aligned}\{\text{preference, pref. deviation}\} \sim\ & \text{AbsPref} \times \ln(\text{OverallFreq}) \\ & + \text{RelFreq} \times \ln(\text{OverallFreq}) + (1 \mid \text{binomial})\end{aligned} \tag{1}$$

Results

Preference The results of the Bayesian mixed-effects regression for preference at the final checkpoint are visualized in Figure 2 (top; for full results, see Table 4, Appendix). The results suggest that there is an effect of relative frequency on ordering preferences across all six models, which is consistent with models' ordering preferences being at least partly

[3]For each statistical analysis, posterior predictive checks suggested adequate model fit. These are presented in the appendix, section Section E.

driven by the distributional statistics of their training data. Interestingly, all models except the C4 1.3B parameter model show an effect of AbsPref (More than 89% of posterior samples are greater than zero for the other 5 models). More specifically, the effect of AbsPref shows a graded pattern across model sizes, with the 125M parameter models showing a clear positive effect with the two smaller models for both datasets showing a qualitatively (and in the case of the C4 1.3B model, quantitatively) weaker effect.

Preference Deviation The results of the Bayesian mixed-effects regression for preference deviation at the final checkpoint are visualized in Figure 2 (bottom; for full results, see Table 5, Appendix). The results for relative frequency are quite mixed across model sizes and corpora. Among the BabyLM models, the 350M model shows a negative effect of relative frequency, while the two other models (125M and 1.3B) show little to no effect, suggesting that the smaller model's ordering preferences are less extreme than training would suggest (i.e., for binomials that occur more in one ordering over the other, the model underestimates the strength of the ordering preference expected by training data). Among the C4 models, the pattern is slightly different: the two smaller models show a negative effect of relative frequency, while the largest shows little evidence of an effect, suggesting that the two smaller models similarly underestimate the strength of the ordering preference. In terms of abstract preferences, the 125M BabyLM model shows a weak positive effect of AbsPref, while the two larger BabyLM models show no evidence of an effect. For the C4 models, the pattern is similar: the C4 125M model shows a weak effect of AbsPref (approximately 89% of posterior samples were greater than zero), while the two larger C4 models show little evidence of an effect.

Taken together, the preference deviation results suggest that while models' ordering preferences are broadly consistent with their training data, the smaller models' preferences are also driven by abstract preferences. These results suggest that smaller models may be more sensitive to abstract ordering preferences even when controlling for frequency.

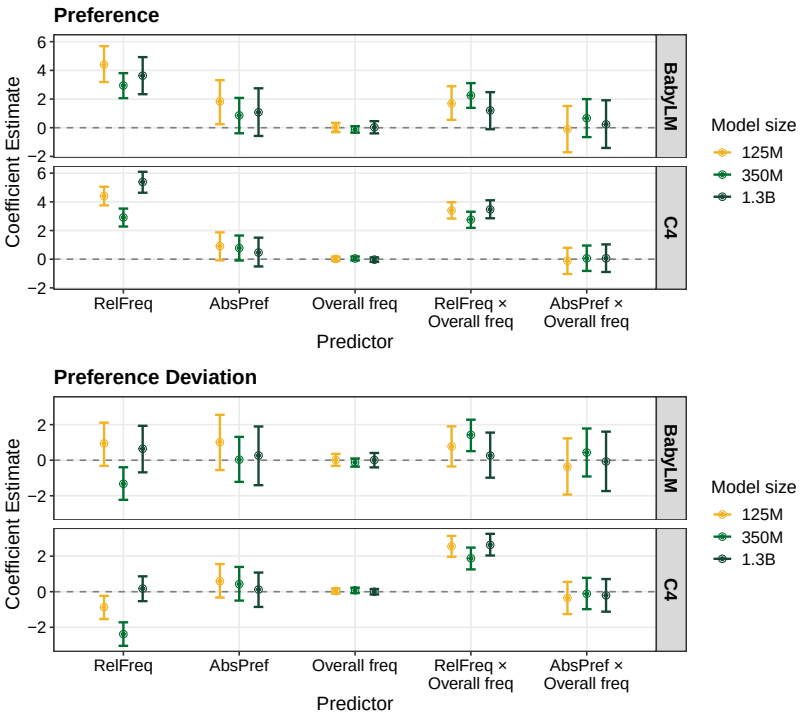

Figure 2: Coefficient estimates at the final training checkpoint for preference (top) and preference deviation (bottom). Error bars show 95% credible intervals.

3.2.2 *Development of Preferences Across Training*

To examine how ordering preferences develop over the course of training, we fit the regression model in Equation 1 separately at 20 evenly-spaced checkpoints throughout training. Notably, the values for overall frequency and relative frequency are calculated at the respective checkpoint. We additionally examine how the overall magnitude of preference deviations changes by computing the **mean absolute preference deviation** across all binomials at each checkpoint, modeled with a log-normal mixed-effects regression predicting absolute preference deviation from training stage interacted with log overall frequency (for full results, see Table 6, Appendix).

Preference and Preference Deviation AbsPref trajectories across training checkpoints are shown in Figure 3 (For a visualization of the other coefficient estimates as a function of training, see Appendix section Section F). Both the smallest and largest babyLM model seem to show an effect of abstract preference which emerges mid-training and remains throughout. Similarly, the smallest C4 model also shows an effect of abstract preference that remains throughout training. Interestingly, the larger C4 model shows a U-shaped effect: an effect of abstract preference emerges early on, but is subsequently reduced by further training. The effect of abstract preference on preference deviation is qualitatively similar, for the BabyLM 1.3B model which shows no evidence of an effect of abstract preference. However, the U-shaped curve is present for the 1.3B C4 model for preference deviation as well. The lack of a U-shaped curve for the BabyLMs suggest that training data diversity may also play a role in the learning of abstract preferences. The coefficient values for the other variables (See Appendix Section Section F) were not particularly surprising: the effect of relative frequency rises throughout training, and for most models that effect is stronger for high-frequency binomials than low-frequency binomials (an effect that emerges throughout training). The only notable exception is the 1.3B parameter model of babyLM, where the effect of relative frequency is stronger for high-frequency binomials in the mid-point of training, but this effect is reduced throughout training, likely due to slight overfitting of the model. Notably, there is no interaction effect between AbsPref and overall frequency, suggesting that models are not relying more on abstract preferences for low-frequency binomials compared to high-frequency binomials.

Mean Absolute Preference Deviation Beyond the direction of preferences, we examined how the overall magnitude of preference deviations changes over training. The mean absolute preference deviation grows throughout training for all six models (Figure 4), indicating that models' preferences deviate increasingly from what relative frequency alone would predict. Among the BabyLM models, this growth is most pronounced for the 1.3B model and weakest for the 350M model. Among the C4 models, all three follow a broadly similar trajectory.

The results of the Bayesian mixed-effects regression for absolute preference deviation are visualized in Figure 5 (for full results, see Table 6, Appendix). The effect of training stage is quite mixed across models. Among the BabyLM models, the 125M and 1.3B models show a positive effect, while the 350M model shows a negative effect. Among the C4 models, the two smaller models show a slightly negative effect while the largest shows a slightly positive effect. In general, the effect of checkpoint appears weaker for the C4 models than for the BabyLM models, suggesting that training data diversity may dampen the growth of preference magnitude (indeed, this is expected following evidence of adaptive partial pooling in transformer models, Kapatsinski, 2026).

The effect of overall frequency is much clearer. All six models show a strong positive effect of overall frequency, indicating that the gap between model preference and the frequency-based prediction is larger for higher-frequency binomials. This pattern is of particular interest given existing work demonstrating frequency-dependent preference extremity in humans, whereby humans develop more extreme ordering preferences for high-frequency binomials (e.g., Houghton, 2025a; Houghton & Morgan, 2024; Morgan & Levy, 2016b). The present results suggest that under certain circumstances language models may show a similar pattern, though a full investigation of this question is left for future research.

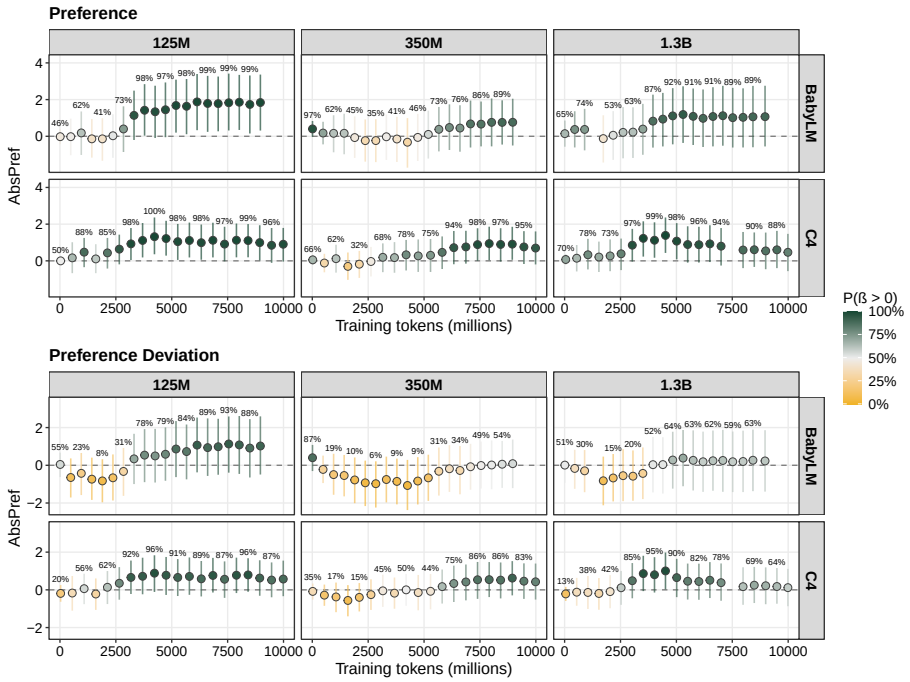

Figure 3: AbsPref coefficient estimates (and 95% CI) at each of 20 sampled training checkpoints for raw preference (top) and preference deviation (bottom) models. Point fill encodes $P(\beta > 0)$.

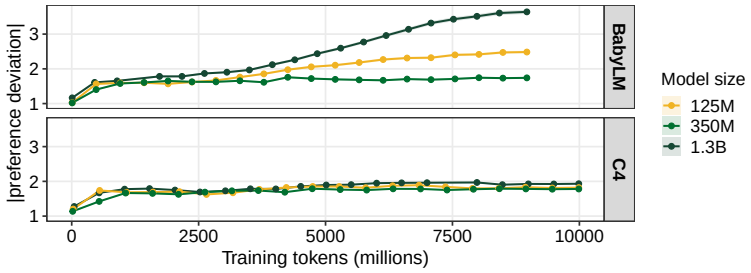

Figure 4: Mean absolute preference deviation across items at each of 20 sampled training checkpoints. Higher values indicate that model preferences deviate more from what relative frequency alone would predict.

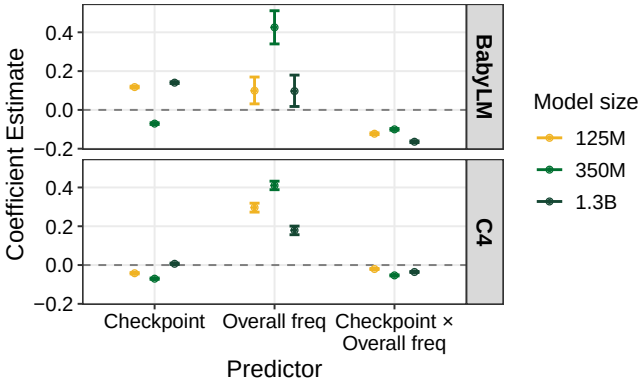

Figure 5: Coefficient estimates for predictors of absolute preference deviation (log-normal likelihood). Coefficients are on the log scale.

4 Discussion and Conclusion

The results of the present study suggest that the models trained here do develop binomial ordering preferences over the course of training, and that these preferences are at least partly driven by the distributional statistics of the training data. Beyond frequency, however, the picture is more nuanced. Smaller models (particularly the 125M models across both corpora) show the most consistent evidence of AbsPref effects, suggesting that they are sensitive to abstract ordering preferences over and above what relative frequency alone would predict. Larger models, by contrast, show weaker or absent AbsPref effects at the final checkpoint, though the trajectory results suggest that these models do show some sensitivity to AbsPref earlier in training, with the effect weakening as training progresses. This trend is more clear in the C4 models, suggesting that data diversity may play a role in this. These results may reflect a tendency for larger models to gradually shift toward relying on distributional frequency as they accumulate more training experience (and diversity), at the expense of abstract knowledge. Finally, as training progresses, models' preferences grow more extreme in a direction broadly consistent with relative frequency, but the absolute gap between model preferences and frequency-based predictions also grows, with the size of this gap modulated by both model size and overall frequency in a manner reminiscent of frequency-dependent preference extremity observed in humans.

Overall, the results demonstrate that both model size and data diversity play a role in the learning of abstract preferences, with evidence that larger models learn abstract preferences early on, but these diminish over further training. On the other hand, smaller models learn the preferences and seem to maintain that knowledge, perhaps due to lacking the parameters to represent each of the binomials distinctly. These results suggest that the lack of abstract preferences found in Houghton et al. (2025) may be due to training dynamics rather than an architectural limitation.

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A The XOR Problem

Consider the XOR problem, in which a model must output 1 for inputs (1,0) and (0,1), but 0 for inputs (0,0) and (1,1). Plotting these four points in two-dimensional input space reveals the difficulty: the two positive cases sit at diagonally opposite corners, with no straight line that separates them from the two negative cases (Figure 6, Panel A). A hidden layer solves this by mapping the four input points into a new representational space defined by the hidden neurons' activations. Each hidden neuron draws its own linear boundary through the input space, assigning each input point a new position in a coordinate space defined by those activation values. To see this concretely, consider two hidden neurons applied to the XOR inputs. The first neuron might learn to fire for any input where $x_1 + x_2 > 0.5$, drawing a diagonal boundary that separates (0,0) from the other three points. The second might learn to fire only when $x_1 + x_2 > 1.5$, drawing a parallel boundary that separates (1,1) from the other three. Each of the four input points now has a pair of activation values, one from each hidden neuron, that describes which side of each boundary it falls on. The input (0,0) activates neither neuron; (1,0) and (0,1) activate the first but not the second; and (1,1) activates both. These three distinct activation signatures relocate the four points into a new space (Figure 6, Panel B) where the positive cases (1,0) and (0,1) are cleanly separated from the negative cases by a single line, a separation that was geometrically impossible in the original input coordinates.

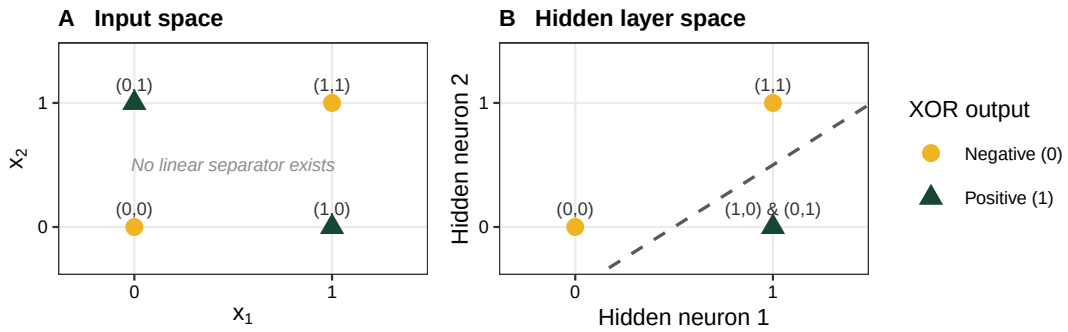

Figure 6: The XOR problem illustrated. **A**: In the original input space, no straight line separates the positive cases (1,0) and (0,1) from the negative cases (0,0) and (1,1). **B**: After applying two hidden neurons (one that fires when $x_1 + x_2 > 0.5$, another when $x_1 + x_2 > 1.5$), the four input points are relocated in hidden activation space. The two positive inputs map to the same point $(1, 0)$, and a single linear boundary (dashed line) now correctly separates positive from negative cases.

B Training Configuration

Corpus	Params	Total tokens	Checkpoints	LR
BabyLM	125M	$\sim$ 9.1B	$\sim$ 910	3×10^{-4}
BabyLM	350M	$\sim$ 9.1B	$\sim$ 910	1×10^{-4}
BabyLM	1.3B	$\sim$ 9.1B	$\sim$ 860	1×10^{-4}
C4	125M	$\sim$ 10B	$\sim$ 1,010	3×10^{-4}
C4	350M	$\sim$ 10B	$\sim$ 1,005	1×10^{-4}
C4	1.3B	$\sim$ 10B	$\sim$ 960	1×10^{-4}

Table 2: Training configuration for all six models. All models use the OPT architecture, a shared BPE tokenizer (vocab 8,192; context 1,024 tokens), AdamW optimization, and a linear LR schedule with 4,000 warmup steps.

C Training Quality

Corpus	Params	Initial PPL	Final PPL
BabyLM	125M	9725	537.50
BabyLM	350M	9470	289.68
BabyLM	1.3B	13078	2557.08
C4	125M	9528	31.63
C4	350M	9352	43.16
C4	1.3B	12275	27.87

Table 3: Perplexity at initialization and final checkpoint for all six models. Perplexity = e^{loss}; initial perplexity is derived from the first logged training step; final perplexity is measured on a held-out validation set (first 5% of BabyLM training split for BabyLM models; official C4 validation split for C4 models). The BabyLM 1.3B model's elevated final perplexity relative to 350M reflects overfitting pressure on a small corpus.

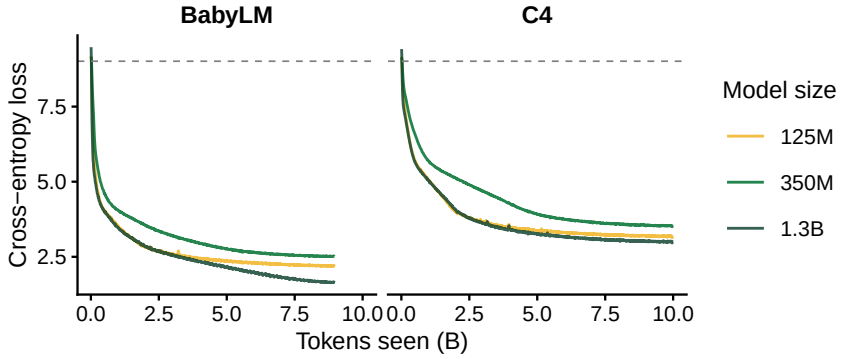

Figure 7: Training loss over the course of training for all six models, separated by corpus. All models start near the theoretical random baseline ($\ln 8192 \approx 9.01$) and converge smoothly; the 1.3B model achieves the lowest final loss in both corpora.

D Mathematical Definitions

D.1 Preference

For a binomial with alphabetically ordered token sequence s_α and non-alphabetically ordered sequence s_n, and a set of $K = 51$ prompt prefixes $\{c_k\}_{k=1}^{K}$, the model's **preference** is:

$$\begin{aligned}\text{preference}_i &= \frac{1}{K}\sum_{k=1}^{K}\Big[\log P_\theta(c_k \circ s_{\alpha,i}) - \log P_\theta(c_k \circ s_{n,i})\Big] \\ &= \frac{1}{K}\sum_{k=1}^{K}\Big[\log P_\theta(s_{\alpha,i} \mid c_k) - \log P_\theta(s_{n,i} \mid c_k)\Big]\end{aligned}$$

where $\log P_\theta(s)$ is the summed token log-probability of sequence s under next-token prediction with model parameters θ, $\circ$ denotes string concatenation, and the equality follows from the chain rule of probability since $\log P_\theta(c_k)$ cancels across the difference.

D.2 Preference Deviation

The **preference deviation** is the difference between a model's preference and the log relative frequency ratio:

$$\text{preference deviation}_i = \text{preference}_i - \ln \frac{c_{\alpha,i}}{c_{n,i}}$$

where $c_{\alpha,i}$ and $c_{n,i}$ are the corpus counts of the alphabetically ordered and non-alphabetically ordered forms of binomial i, respectively. A value of zero indicates that the model's preference exactly matches what relative frequency predicts; positive values indicate a stronger preference for the alpha-ordered form than the corpus would suggest.

D.3 Relative Frequency

RelFreq is the proportion of corpus occurrences in the alphabetically ordered form, centered:

$$\text{RelFreq}_i = P(\text{alpha})_i - 0.5, \quad \text{where} \quad P(\text{alpha})_i = \frac{c_{\alpha,i}}{c_{\alpha,i} + c_{n,i}}$$

where $c_{\alpha,i}$ and $c_{n,i}$ are corpus counts of the alpha-ordered and non-alphabetical forms of binomial i.

E Posterior Predictive Checks

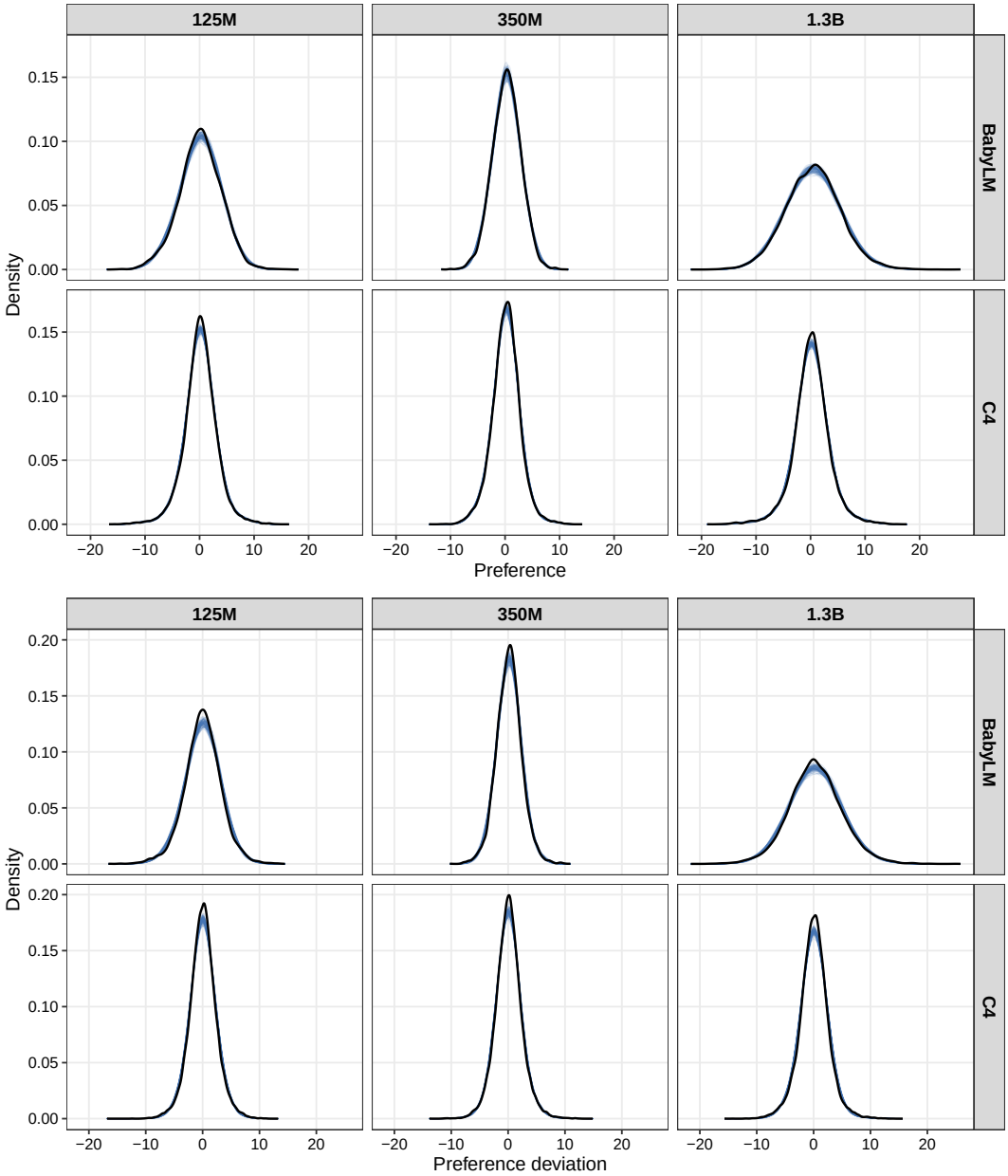

Figure 8: Posterior predictive checks for the preference model (top) and preference deviation model (bottom) at the final training checkpoint. The dark line shows the observed outcome density; light lines show 100 replicated datasets drawn from the posterior predictive distribution.

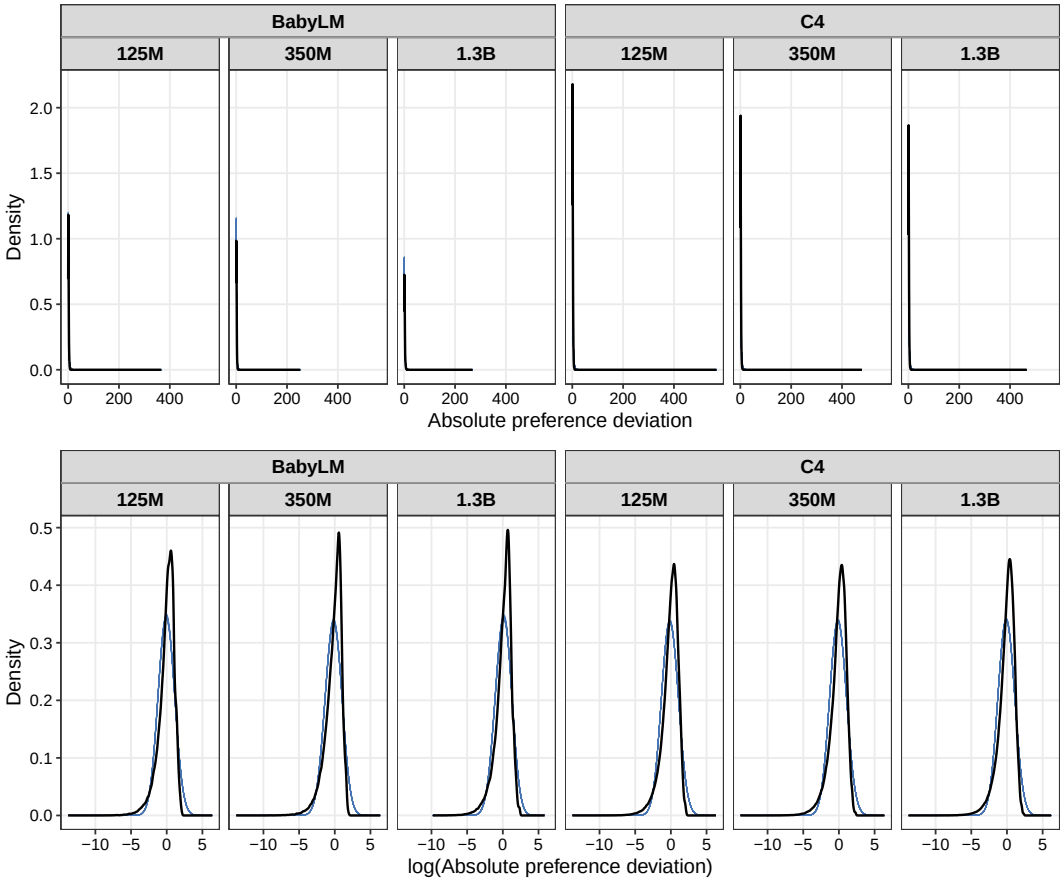

Figure 9: Posterior predictive check for the absolute preference deviation model (log-normal likelihood). Top panels show densities on the raw scale; bottom panels show densities after log-transforming the outcome, which spreads out the distribution and makes fit easier to assess. In both cases the dark line shows the observed outcome density and light lines show 100 replicated datasets drawn from the posterior predictive distribution.

F Coefficient Trajectories Across Training

Each figure below follows the same format as Figure 3. Point fill encodes $P(\beta > 0)$; gold indicates $P(\beta > 0) < 0.5$ (evidence against a positive effect) and green indicates $P(\beta > 0) > 0.5$ (evidence for a positive effect).

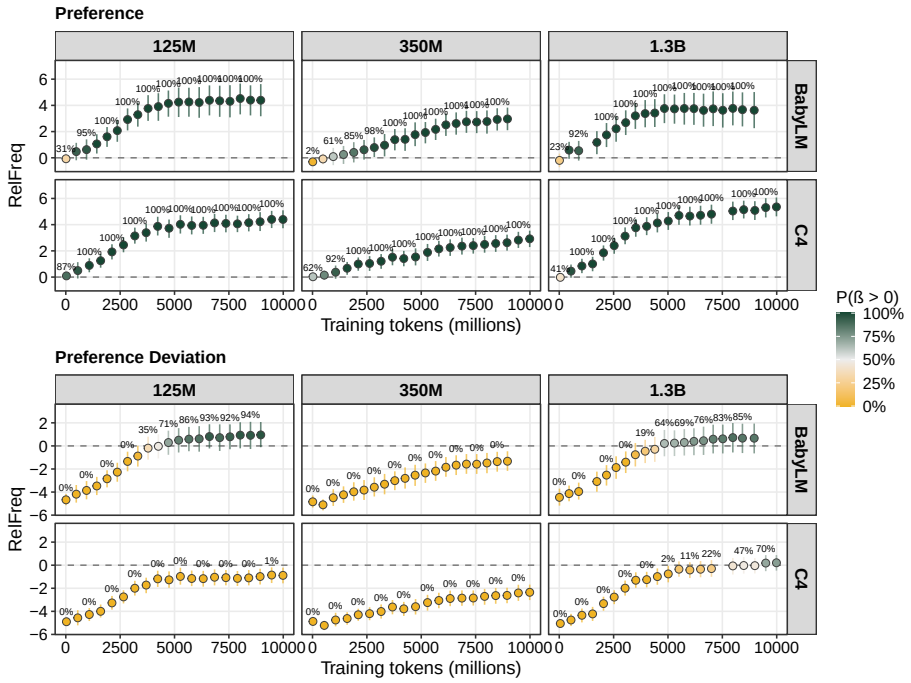

Figure 10: RelFreq coefficient trajectory across training checkpoints.

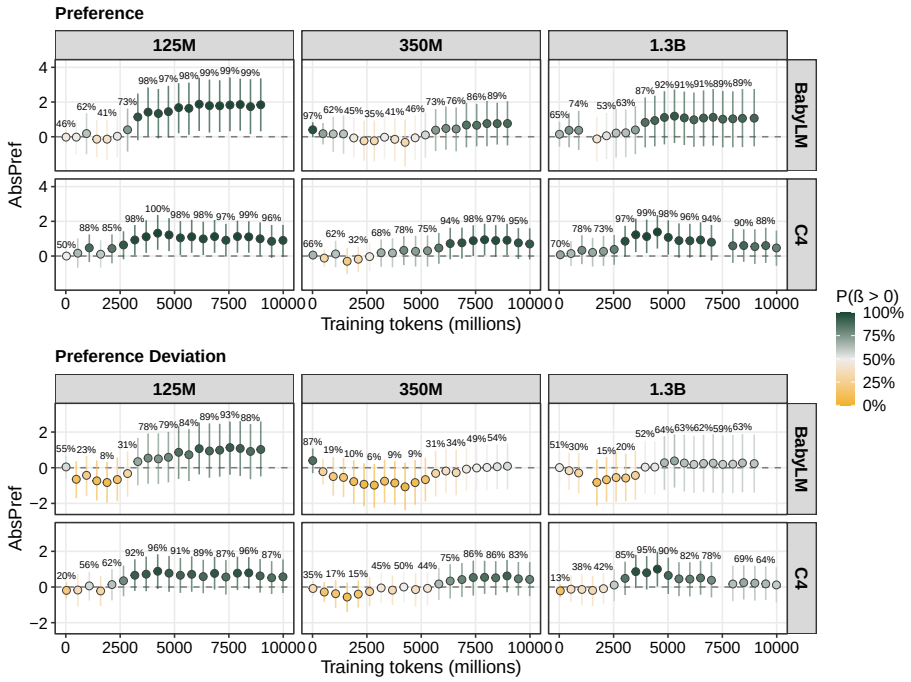

Figure 11: AbsPref coefficient trajectory across training checkpoints.

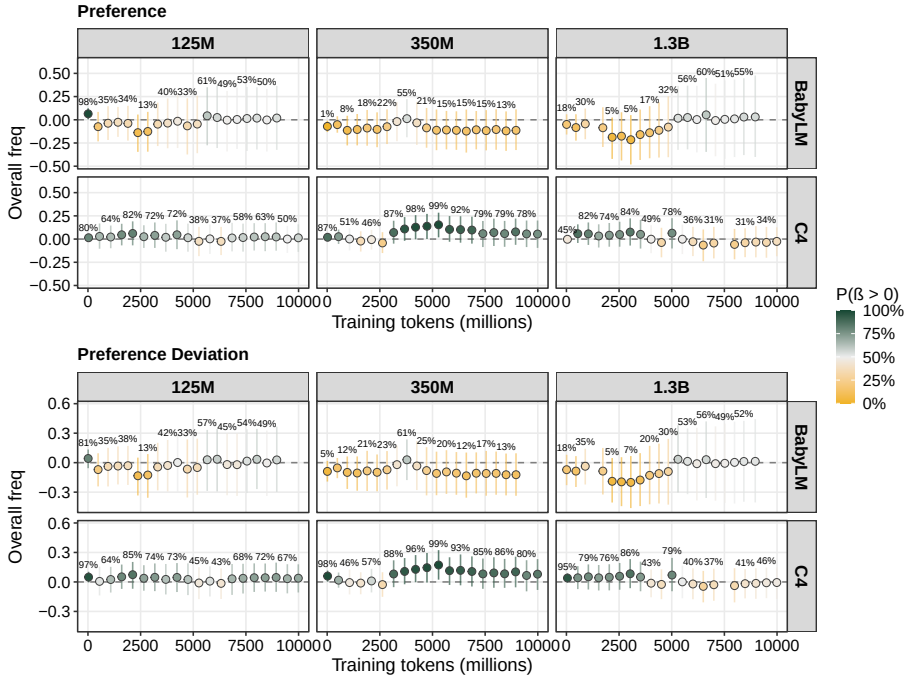

Figure 12: Overall frequency coefficient trajectory across training checkpoints.

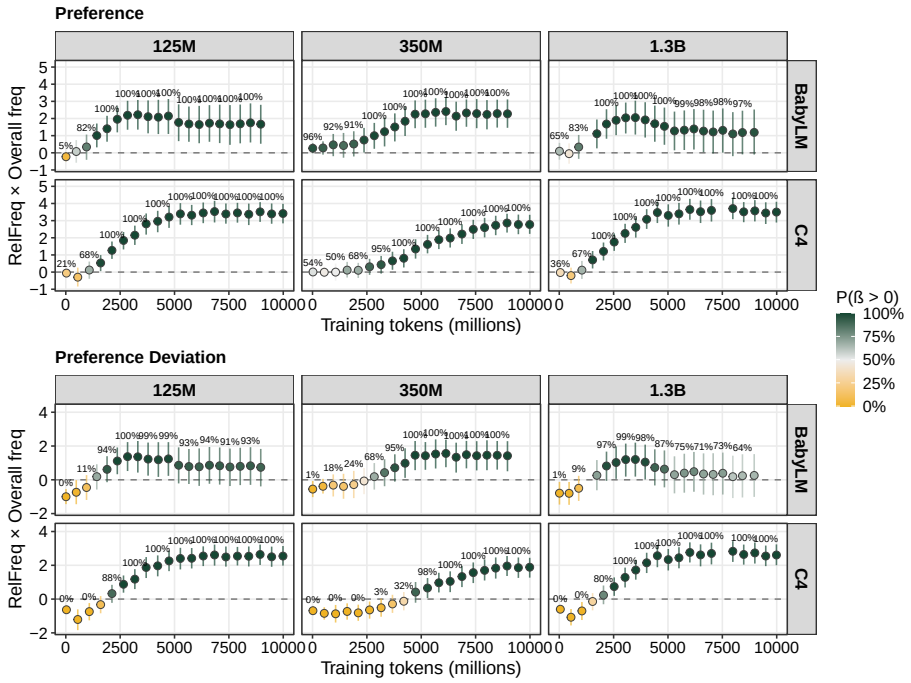

Figure 13: RelFreq × Overall frequency interaction coefficient trajectory across training checkpoints.

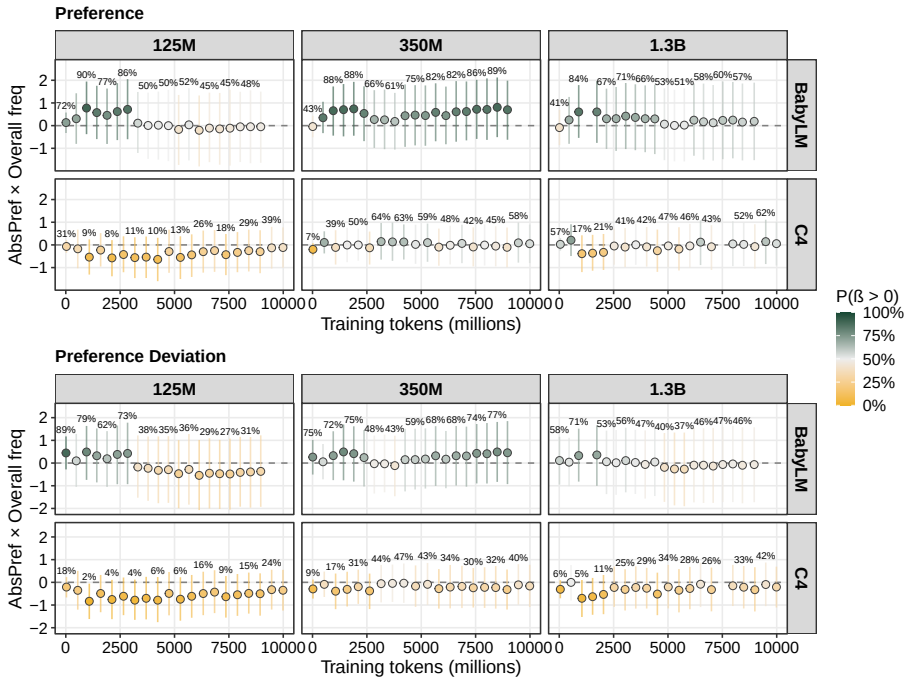

Figure 14: AbsPref × Overall frequency interaction coefficient trajectory across training checkpoints.

G Posterior Coefficient Tables

G.1 Preference: Final Checkpoint

Corpus	Size	Term	Mean	SD	2.5%	97.5%	$P(\beta > 0)$
BabyLM	125M	AbsPref	1.835	0.790	0.243	3.313	0.990
BabyLM	125M	AbsPref × Overall freq	-0.105	0.827	-1.713	1.514	0.449
BabyLM	125M	Overall freq (ln total)	0.024	0.170	-0.312	0.339	0.556
BabyLM	125M	RelFreq × Overall freq	1.688	0.587	0.553	2.892	0.998
BabyLM	125M	RelFreq (P(alpha)-0.5)	4.404	0.638	3.185	5.687	1.000
BabyLM	350M	AbsPref	0.863	0.643	-0.388	2.080	0.910
BabyLM	350M	AbsPref × Overall freq	0.664	0.675	-0.655	1.996	0.838
BabyLM	350M	Overall freq (ln total)	-0.120	0.117	-0.344	0.106	0.152
BabyLM	350M	RelFreq × Overall freq	2.254	0.442	1.385	3.112	1.000
BabyLM	350M	RelFreq (P(alpha)-0.5)	2.945	0.447	2.062	3.806	1.000
BabyLM	1.3B	AbsPref	1.078	0.848	-0.573	2.750	0.898
BabyLM	1.3B	AbsPref × Overall freq	0.234	0.854	-1.412	1.914	0.608
BabyLM	1.3B	Overall freq (ln total)	0.029	0.217	-0.394	0.458	0.554
BabyLM	1.3B	RelFreq × Overall freq	1.205	0.659	-0.106	2.480	0.966
BabyLM	1.3B	RelFreq (P(alpha)-0.5)	3.632	0.666	2.337	4.924	1.000
C4	125M	AbsPref	0.917	0.493	-0.080	1.872	0.969
C4	125M	AbsPref × Overall freq	-0.113	0.465	-1.034	0.796	0.404
C4	125M	Overall freq (ln total)	0.021	0.078	-0.131	0.180	0.607
C4	125M	RelFreq × Overall freq	3.401	0.294	2.835	3.972	1.000
C4	125M	RelFreq (P(alpha)-0.5)	4.406	0.336	3.754	5.054	1.000
C4	350M	AbsPref	0.777	0.452	-0.091	1.652	0.957
C4	350M	AbsPref × Overall freq	0.061	0.452	-0.817	0.953	0.553
C4	350M	Overall freq (ln total)	0.050	0.071	-0.085	0.190	0.761
C4	350M	RelFreq × Overall freq	2.760	0.287	2.185	3.309	1.000
C4	350M	RelFreq (P(alpha)-0.5)	2.913	0.319	2.279	3.530	1.000
C4	1.3B	AbsPref	0.466	0.509	-0.505	1.502	0.820
C4	1.3B	AbsPref × Overall freq	0.063	0.487	-0.892	1.033	0.552
C4	1.3B	Overall freq (ln total)	-0.031	0.080	-0.186	0.124	0.351
C4	1.3B	RelFreq × Overall freq	3.472	0.323	2.855	4.109	1.000
C4	1.3B	RelFreq (P(alpha)-0.5)	5.389	0.375	4.635	6.097	1.000

Table 4: Coefficient estimates for preference at the final checkpoint. Mean = posterior mean; SD = posterior SD; 2.5%/97.5% = 95% credible interval bounds.

G.2 Preference Deviation: Final Checkpoint

Corpus	Size	Term	Mean	SD	2.5%	97.5%	$P(\beta > 0)$
BabyLM	125M	AbsPref	1.013	0.796	-0.551	2.555	0.898
BabyLM	125M	AbsPref × Overall freq	-0.364	0.808	-1.931	1.230	0.326
BabyLM	125M	Overall freq (ln total)	0.020	0.171	-0.316	0.353	0.546
BabyLM	125M	RelFreq × Overall freq	0.775	0.579	-0.350	1.906	0.910
BabyLM	125M	RelFreq (P(alpha)-0.5)	0.948	0.612	-0.316	2.113	0.939
BabyLM	350M	AbsPref	0.033	0.642	-1.218	1.312	0.521
BabyLM	350M	AbsPref × Overall freq	0.436	0.683	-0.912	1.785	0.738
BabyLM	350M	Overall freq (ln total)	-0.119	0.116	-0.362	0.096	0.153
BabyLM	350M	RelFreq × Overall freq	1.427	0.444	0.511	2.281	0.999
BabyLM	350M	RelFreq (P(alpha)-0.5)	-1.325	0.469	-2.228	-0.395	0.002
BabyLM	1.3B	AbsPref	0.265	0.842	-1.404	1.900	0.624
BabyLM	1.3B	AbsPref × Overall freq	-0.069	0.848	-1.733	1.608	0.468
BabyLM	1.3B	Overall freq (ln total)	0.013	0.209	-0.402	0.408	0.525
BabyLM	1.3B	RelFreq × Overall freq	0.262	0.651	-0.982	1.555	0.656
BabyLM	1.3B	RelFreq (P(alpha)-0.5)	0.645	0.663	-0.682	1.936	0.835
C4	125M	AbsPref	0.595	0.475	-0.328	1.553	0.895
C4	125M	AbsPref × Overall freq	-0.351	0.462	-1.256	0.552	0.224
C4	125M	Overall freq (ln total)	0.039	0.073	-0.103	0.183	0.700
C4	125M	RelFreq × Overall freq	2.557	0.304	1.966	3.139	1.000
C4	125M	RelFreq (P(alpha)-0.5)	-0.874	0.334	-1.537	-0.230	0.004
C4	350M	AbsPref	0.430	0.478	-0.500	1.394	0.816
C4	350M	AbsPref × Overall freq	-0.115	0.456	-0.980	0.780	0.401
C4	350M	Overall freq (ln total)	0.079	0.078	-0.076	0.226	0.847
C4	350M	RelFreq × Overall freq	1.881	0.313	1.256	2.482	1.000
C4	350M	RelFreq (P(alpha)-0.5)	-2.385	0.340	-3.043	-1.721	0.000
C4	1.3B	AbsPref	0.125	0.488	-0.855	1.077	0.601
C4	1.3B	AbsPref × Overall freq	-0.207	0.466	-1.123	0.713	0.328
C4	1.3B	Overall freq (ln total)	-0.002	0.077	-0.153	0.151	0.489
C4	1.3B	RelFreq × Overall freq	2.632	0.312	2.038	3.255	1.000
C4	1.3B	RelFreq (P(alpha)-0.5)	0.175	0.355	-0.530	0.865	0.689

Table 5: Coefficient estimates for preference deviation at the final checkpoint.

G.3 Absolute Preference Deviation (Log-Normal)

Corpus	Size	Term	Mean	SD	2.5%	97.5%	$P(\beta > 0)$
BabyLM	125M	Checkpoint × Overall freq	-0.123	0.002	-0.127	-0.118	0.000
BabyLM	125M	Checkpoint (tokens, z)	0.118	0.002	0.113	0.122	1.000
BabyLM	125M	Overall freq (ln total, z within ckpt)	0.099	0.037	0.031	0.169	0.997
BabyLM	350M	Checkpoint × Overall freq	-0.100	0.002	-0.105	-0.096	0.000
BabyLM	350M	Checkpoint (tokens, z)	-0.071	0.002	-0.075	-0.066	0.000
BabyLM	350M	Overall freq (ln total, z within ckpt)	0.425	0.044	0.340	0.511	1.000
BabyLM	1.3B	Checkpoint × Overall freq	-0.164	0.002	-0.169	-0.160	0.000
BabyLM	1.3B	Checkpoint (tokens, z)	0.140	0.002	0.136	0.145	1.000
BabyLM	1.3B	Overall freq (ln total, z within ckpt)	0.097	0.040	0.018	0.179	0.992
C4	125M	Checkpoint × Overall freq	-0.020	0.001	-0.023	-0.018	0.000
C4	125M	Checkpoint (tokens, z)	-0.042	0.002	-0.046	-0.039	0.000
C4	125M	Overall freq (ln total, z within ckpt)	0.297	0.012	0.273	0.319	1.000
C4	350M	Checkpoint × Overall freq	-0.053	0.001	-0.056	-0.050	0.000
C4	350M	Checkpoint (tokens, z)	-0.070	0.002	-0.074	-0.067	0.000
C4	350M	Overall freq (ln total, z within ckpt)	0.410	0.011	0.388	0.432	1.000
C4	1.3B	Checkpoint × Overall freq	-0.036	0.001	-0.039	-0.033	0.000
C4	1.3B	Checkpoint (tokens, z)	0.006	0.002	0.003	0.010	1.000
C4	1.3B	Overall freq (ln total, z within ckpt)	0.179	0.011	0.156	0.201	1.000

Table 6: Coefficient estimates for absolute preference deviation (log-normal likelihood). Coefficients are on the log scale. Mean = posterior mean; SD = posterior SD; 2.5%/97.5% = 95% credible interval bounds.